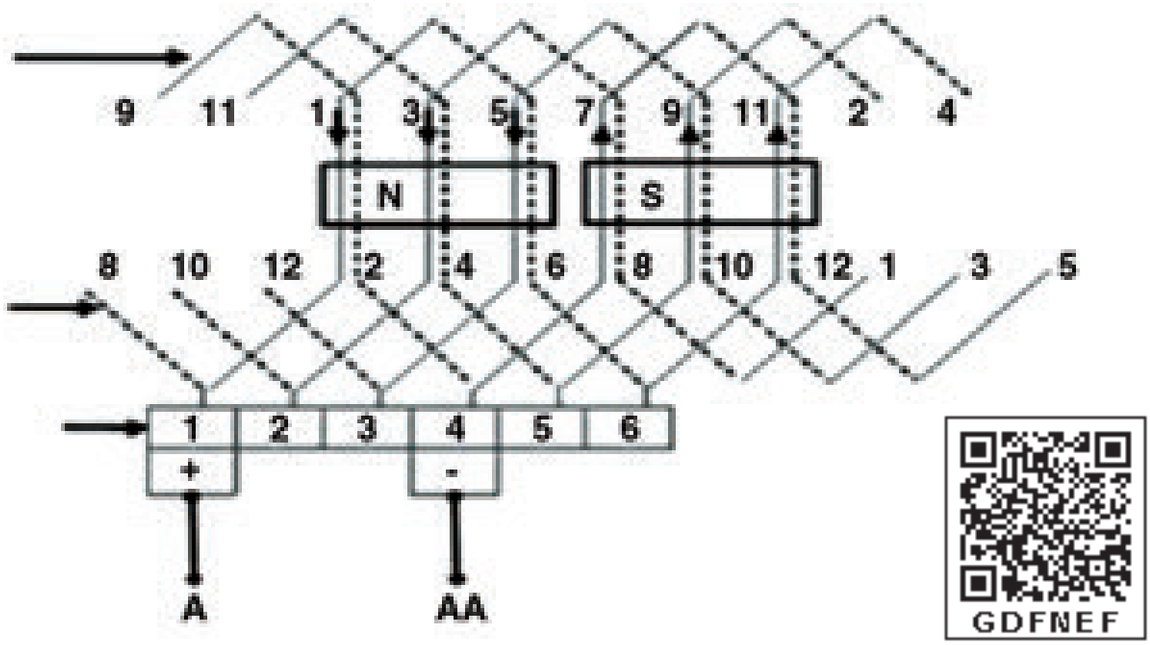


Fig. 9.11 2 Poles 6 slots simplex wave winding

9.6 AC windings

The winding made in the AC machines are called AC windings.

9.6.1 Types of AC windings

Generally three types of winding methods were carried out in AC machines. They are

i. Lap winding
ii. Wave winding and
iii. Concentric winding

9.6.2 Lap and wave winding

Lap windings are generally used in low voltage and high current electrical machines. But the wave windings are used in high voltage and low current electrical machines.

9.6.3 Concentric winding

The centre of all the coils in a coil group confined to a constant centre as shown in the figure 9.12 is called concentric winding.

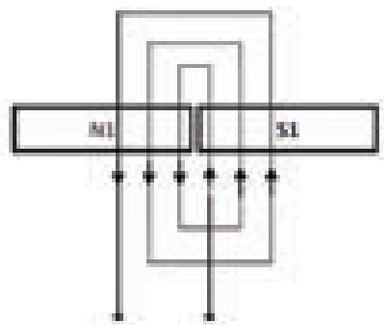


Fig. 9.12 Concentric winding

9.6.4 Coil group

Coil group consists of one coil or more than one coil connected in series.

When we carried out AC winding calculation, the number of coils per pole should be calculated. Then the coils in a pole should be separated for three phases also. Thus the coils per pole per phase is made into a coil group. By making coil groups, the number of end connections can be reduced and hence we can prevent short circuits in the end connection areas.

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